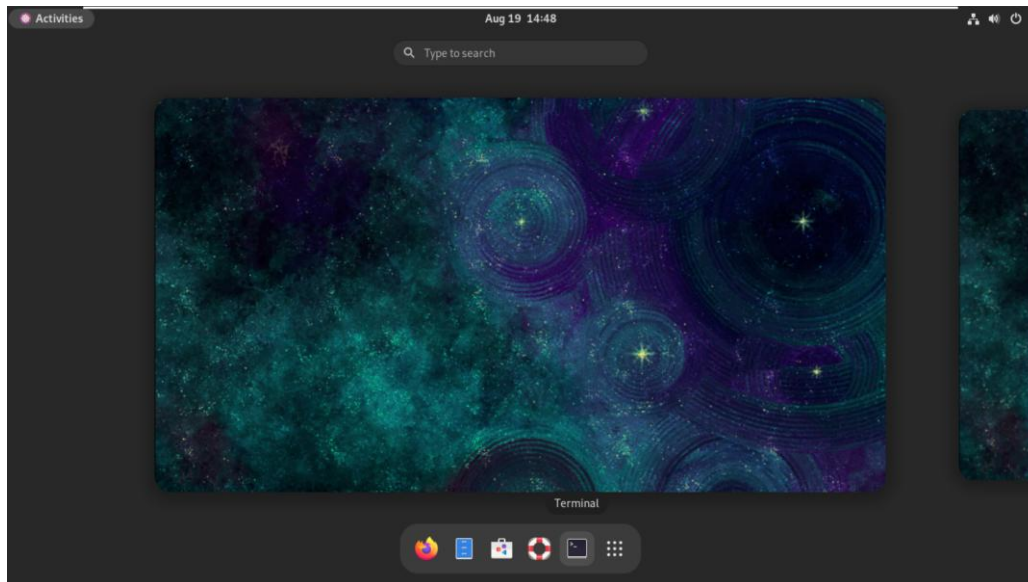


Linux Basic commands-27

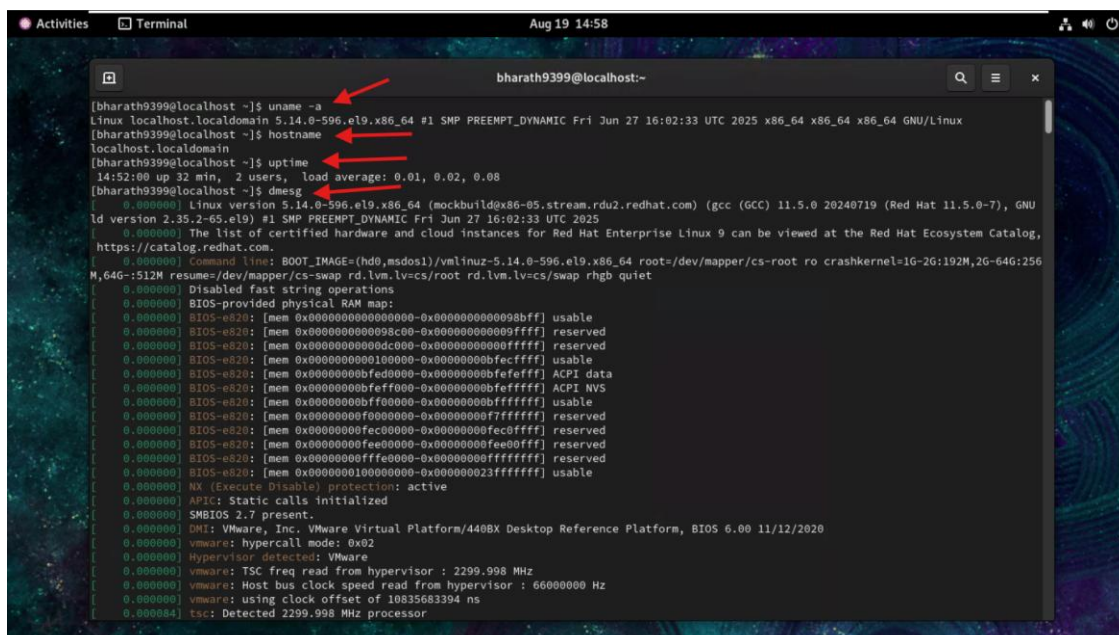
After installing a linux system in VMware we can open the command terminal

Now open terminal



Here are some of the command line commands

- `uname -a` – Shows all system information (kernel name, version, architecture).
- `hostname` – Displays the system's hostname.
- `uptime` – Shows how long the system has been running.
- `dmesg` – Prints kernel ring buffer messages (boot and hardware logs).



- arch – Shows machine architecture (like x86_64).
- date – Prints current date and time.
- whoami – Shows the current logged-in user.
- uname – Prints the kernel name.
- uname -r – Displays the kernel version.
- uname -a – Shows full system info again.
- cal – Displays a calendar.
- last reboot – Shows system reboot history.
- timedatectl – Displays or sets system date, time, and timezone.
- w – Shows who is logged in and their processes.

```

[ 239.955627] input: VMware DnD UInput pointer as /devices/virtual/input/input7
[bharath9399@localhost ~]$ arch
x86_64
[bharath9399@localhost ~]$ date
Tuesday 19 August 2025 03:01:06 PM IST
[bharath9399@localhost ~]$ whoami
bharath9399
[bharath9399@localhost ~]$ uname
Linux
[bharath9399@localhost ~]$ uname -r
5.14.0-596.el9.x86_64
[bharath9399@localhost ~]$ cal
      August 2025
Su Mo Tu We Th Fr Sa
                1  2
 3  4  5  6  7  8  9
10 11 12 13 14 15 16
17 18 19 20 21 22 23
24 25 26 27 28 29 30
31
[bharath9399@localhost ~]$ last reboot
reboot    system boot  5.14.0-596.el9.x Tue Aug 19 14:19    still running

wtmp begins Tue Aug 19 14:19:24 2025
[bharath9399@localhost ~]$ timedatectl
      Local time: Tue 2025-08-19 15:04:50 IST
      Universal time: Tue 2025-08-19 09:34:50 UTC
      RTC time: Tue 2025-08-19 09:34:50
      Time zone: Asia/Kolkata (IST, +0530)
System clock synchronized: yes
      NTP service: active
      RTC in local TZ: no
[bharath9399@localhost ~]$ w
15:07:30 up 48 min,  2 users,  load average: 0.00, 0.00, 0.02
USER    TTY          LOGIN@  IDLE   JCPU   PCPU WHAT
bharath9 seat0        14:22    0.00s  0.00s  0.00s /usr/libexec/gdm-wayland-session --register-session gnome-session
bharath9 tty2        14:22   48:03  0.05s  0.05s /usr/libexec/gnome-session-binary
[bharath9399@localhost ~]$

```

- ps -t tty2 – Displays processes running on terminal tty2.
- lshw – Lists detailed hardware information.

```

Aug 19 15:12
bharath9399@localhost:~

[bharath9399@localhost ~]$ ps -t tty2
  PID TTY          TIME CMD
 5768 tty2      00:00:00 gdm-wayland-ses
 5776 tty2      00:00:00 gnome-session-b
[bharath9399@localhost ~]$ lshw
WARNING: you should run this program as super-user.
localhost.localdomain
  description: Computer
    width: 64 bits
    capabilities: vsscall32
*-core
  description: Motherboard
    physical id: 0
*-memory
  description: System memory
    physical id: 0
    size: 8GiB
*-cpu
  product: Intel(R) Xeon(R) CPU E5-2670 v3 @ 2.30GHz
  vendor: Intel Corp.
  physical id: 1
  bus info: cpu@0
  version: 6.63.2
  width: 64 bits
  capabilities: fpu_exception wp vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cm
scall nx pdpe1gb rdtscp x86-64 constant_tsc arch_perfmon nopl xtopology tsc_reliable nonstop_tsc cpu
16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor la
tsc_adjust bmi1 avx2 smep bmi2 invpcid xsaveopt arat md_clear flush_lld arch_capabilities
  configuration: microcode=73
*-pci

```

- lsblk – Lists block devices like disks and partitions.
- df -h – Shows disk space usage in human-readable form.
- df -i – Shows inode usage of file systems.
- du -s – Displays total disk usage of a directory.
- du -sh – Shows disk usage in human-readable format.
- du -sh /var/log – Shows size of /var/log directory.

```

Aug 19 15:27
bharath9399@localhost:~

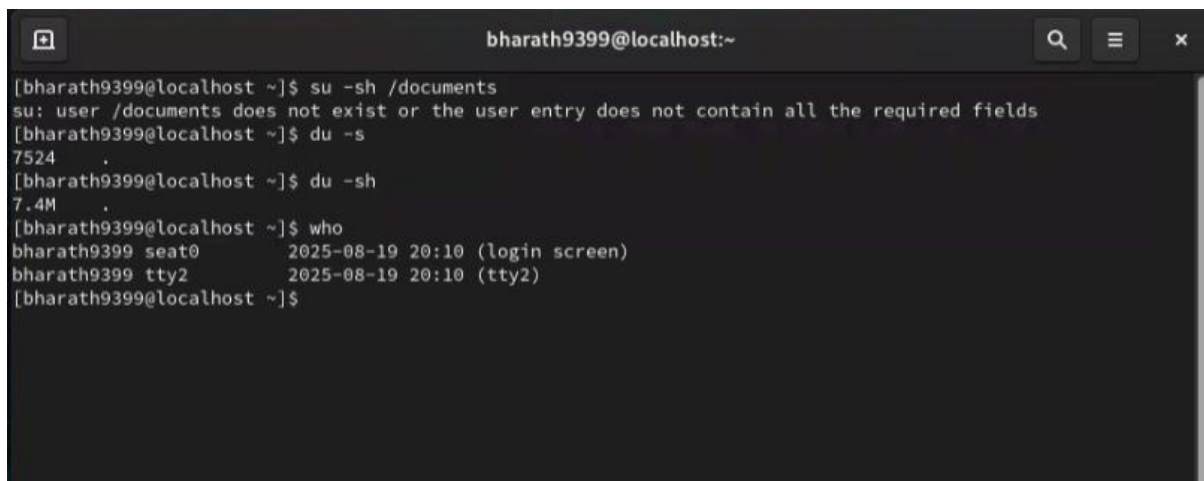
WARNING: output may be incomplete or inaccurate, you should run this program as super-user.
[bharath9399@localhost ~]$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sr0          11:0    1 1024M  0 rom
nvme0n1      259:0    0   80G  0 disk
├─nvme0n1p1 259:1    0    1G  0 part /boot
├─nvme0n1p2 259:2    0    79G  0 part
├─└─cs-root 253:0    0 47.8G  0 lvm  /
├─└─cs-swap 253:1    0  7.8G  0 lvm  [SWAP]
└─└─cs-home 253:2    0 23.3G  0 lvm  /home
[bharath9399@localhost ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs         4.0M   0  4.0M   0% /dev
tmpfs            3.6G   0  3.6G   0% /dev/shm
tmpfs           1.5G  12M  1.5G   1% /run
/dev/mapper/cs-root 48G  5.9G   42G  13% /
/dev/nvme0n1p1 960M 373M 588M 39% /boot
/dev/mapper/cs-home 24G 207M   24G   1% /home
tmpfs           765M 120K 765M   1% /run/user/1000
[bharath9399@localhost ~]$ df -i
Filesystem      Inodes   IUsed   IFree IUse% Mounted on
devtmpfs       970688    468 970220    1% /dev
tmpfs          979048     2 979046    1% /dev/shm
tmpfs         819200  1215 817985    1% /run
/dev/mapper/cs-root 25665472 133375 24932097    1% /
/dev/nvme0n1p1 524288    360 523928    1% /boot
/dev/mapper/cs-home 12236800 284 12236516    1% /home
tmpfs         195809    138 195671    1% /run/user/1000
[bharath9399@localhost ~]$ du -s
7488
[bharath9399@localhost ~]$ du -sh
7.4M
[bharath9399@localhost ~]$ du -sh /var/log
du: cannot read directory '/var/log/private': Permission denied
du: cannot read directory '/var/log/samba': Permission denied
du: cannot read directory '/var/log/audit': Permission denied
du: cannot read directory '/var/log/sss': Permission denied
du: cannot read directory '/var/log/speech-dispatcher': Permission denied
du: cannot read directory '/var/log/gdm': Permission denied

```

- `du -sh /home` – Shows size of /home directory.
- `du -sh /home/*` – Shows size of each user directory under /home.

```
[bharath9399@localhost ~]$ du -sh /home/*
7.5M    /home/bharath9399
[bharath9399@localhost ~]$ du -sh /etc
du: cannot read directory '/etc/lvm/devices': Permission denied
du: cannot read directory '/etc/lvm/archive': Permission denied
du: cannot read directory '/etc/lvm/backup': Permission denied
du: cannot read directory '/etc/lvm/cache': Permission denied
du: cannot read directory '/etc/pki/rsyslog': Permission denied
du: cannot read directory '/etc/sos/cleaner': Permission denied
du: cannot read directory '/etc/nftables': Permission denied
du: cannot read directory '/etc/audit': Permission denied
du: cannot read directory '/etc/sss': Permission denied
du: cannot read directory '/etc/cups/ssl': Permission denied
du: cannot read directory '/etc/sudoers.d': Permission denied
du: cannot read directory '/etc/polkit-1/rules.d': Permission denied
du: cannot read directory '/etc/polkit-1/localauthority': Permission denied
du: cannot read directory '/etc/grub.d': Permission denied
du: cannot read directory '/etc/ssh/ssh_config.d': Permission denied
du: cannot read directory '/etc/firewalld': Permission denied
28M     /etc
[bharath9399@localhost ~]$
```

- `su -sh /documents` – Invalid syntax, but looks like intended to switch user and check /documents.
- `su -sh /etc` – Same issue, but intent was disk usage of /etc.
- `du -sh /etc` – Shows size of /etc directory.
- `ls` – Lists files and directories.
- `pwd` – Prints the current working directory.
- `fdisk -l` – Lists partition tables of all disks.



A terminal window titled 'bharath9399@localhost:~' with search, menu, and close icons in the title bar. The terminal shows the following commands and output:

```
[bharath9399@localhost ~]$ su -sh /documents
su: user /documents does not exist or the user entry does not contain all the required fields
[bharath9399@localhost ~]$ du -s
7524    .
[bharath9399@localhost ~]$ du -sh
7.4M    .
[bharath9399@localhost ~]$ who
bharath9399  seat0          2025-08-19 20:10 (login screen)
bharath9399  tty2            2025-08-19 20:10 (tty2)
[bharath9399@localhost ~]$
```

- `sudo su` – Switches to root user.

```
[bharath9399@localhost ~]$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
[bharath9399@localhost ~]$ pwd
/home/bharath9399
[bharath9399@localhost ~]$ sudo su

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

    #1) Respect the privacy of others.
    #2) Think before you type.
    #3) With great power comes great responsibility.

[sudo] password for bharath9399:
[root@localhost bharath9399]# fdisk -l
Disk /dev/nvme0n1: 80 GiB, 85899345920 bytes, 167772160 sectors
Disk model: VMware Virtual NVMe Disk
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x1fae7b8c

Device            Boot    Start        End    Sectors  Size Id Type
/dev/nvme0n1p1    *                2048    2099199    2097152    1G 83 Linux
/dev/nvme0n1p2                2099200    167772159    165672960    79G 8e Linux LVM

Disk /dev/mapper/cs-root: 47.81 GiB, 51334086656 bytes, 100261888 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/mapper/cs-swap: 7.85 GiB, 8426356736 bytes, 16457728 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/mapper/cs-home: 23.34 GiB, 25060966400 bytes, 48947200 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
[root@localhost bharath9399]#
```

- `who` – Shows logged-in users.
- `exit` – Ends shell session.
- `pvs` – Displays physical volumes in LVM.
- `vgs` – Displays volume groups in LVM.
- `lvs` – Displays logical volumes in LVM.
- `pvdisplay` – Detailed info about physical volumes.
- `vgdisplay` – Detailed info about volume groups.
- `lvdisplay` – Detailed info about logical volumes.


```

[root@localhost bharath9399]# pvs
PV                VG Fmt Attr PSize  PFree
/dev/nvme0n1p2   cs  lvm2 a--  <79.00g    0
[root@localhost bharath9399]# vgs
VG #PV #LV #SN Attr   VSize  VFree
cs    1   3   0 wz--n- <79.00g    0
[root@localhost bharath9399]# lvs
LV      VG Attr      LSize   Pool Origin Data%  Meta%   Move Log Cpy%Sync Convert
home    cs -wi-ao---- <23.34g
root    cs -wi-ao---- <47.81g
swap    cs -wi-ao----  <7.85g
[root@localhost bharath9399]# pvdisplay
--- Physical volume ---
PV Name           /dev/nvme0n1p2
VG Name           cs
PV Size           <79.00 GiB / not usable 3.00 MiB
Allocatable       yes (but full)
PE Size           4.00 MiB
Total PE          20223
Free PE           0
Allocated PE      20223
PV UUID           eZcj24-MdDY-TZTA-s73F-XL2s-1Rh6-KQVWDF

[root@localhost bharath9399]# vgdisplay
--- Volume group ---
VG Name           cs
System ID
Format            lvm2
Metadata Areas    1
Metadata Sequence No 4
VG Access         read/write
VG Status         resizable
MAX LV            0
Cur LV           3
Open LV           3
Max PV            0
Cur PV           1
Act PV            1
VG Size           <79.00 GiB
PE Size           4.00 MiB
Total PE          20223
Alloc PE / Size   20223 / <79.00 GiB
Free PE / Size    0 / 0
VG UUID           vAncgr-Gzy5-FWpI-deCx-T239-fsu9-4Kckev

```

```

[root@localhost bharath9399]# lvdisplay
--- Logical volume ---
LV Path           /dev/cs/swap
LV Name           swap
VG Name           cs
LV UUID           PfmjSS-boZl-CH3c-m5oA-6mmq-n0hS-dvj9fh
LV Write Access   read/write
LV Creation host, time localhost.localdomain, 2025-08-19 13:25:23 +0530
LV Status         available
# open            2
LV Size           <7.85 GiB
Current LE        2009
Segments          1
Allocation        inherit
Read ahead sectors auto
- currently set to 256
Block device      253:1

--- Logical volume ---
LV Path           /dev/cs/home
LV Name           home
VG Name           cs
LV UUID           HyeH6E-uS0e-ad7y-mEMu-KWlp-OTYc-Q6WpxU
LV Write Access   read/write
LV Creation host, time localhost.localdomain, 2025-08-19 13:25:23 +0530
LV Status         available
# open            1
LV Size           <23.34 GiB
Current LE        5975
Segments          1
Allocation        inherit
Read ahead sectors auto
- currently set to 256
Block device      253:2

--- Logical volume ---
LV Path           /dev/cs/root
LV Name           root
VG Name           cs
LV UUID           8SLCkn-CPyV-565m-QpgQ-2iLi-H912-o1NX2h
LV Write Access   read/write
LV Creation host, time localhost.localdomain, 2025-08-19 13:25:24 +0530
LV Status         available
# open            1
LV Size           <47.81 GiB
Current LE        12239
Segments          1
Allocation        inherit
Read ahead sectors auto
- currently set to 256
Block device      253:0

```

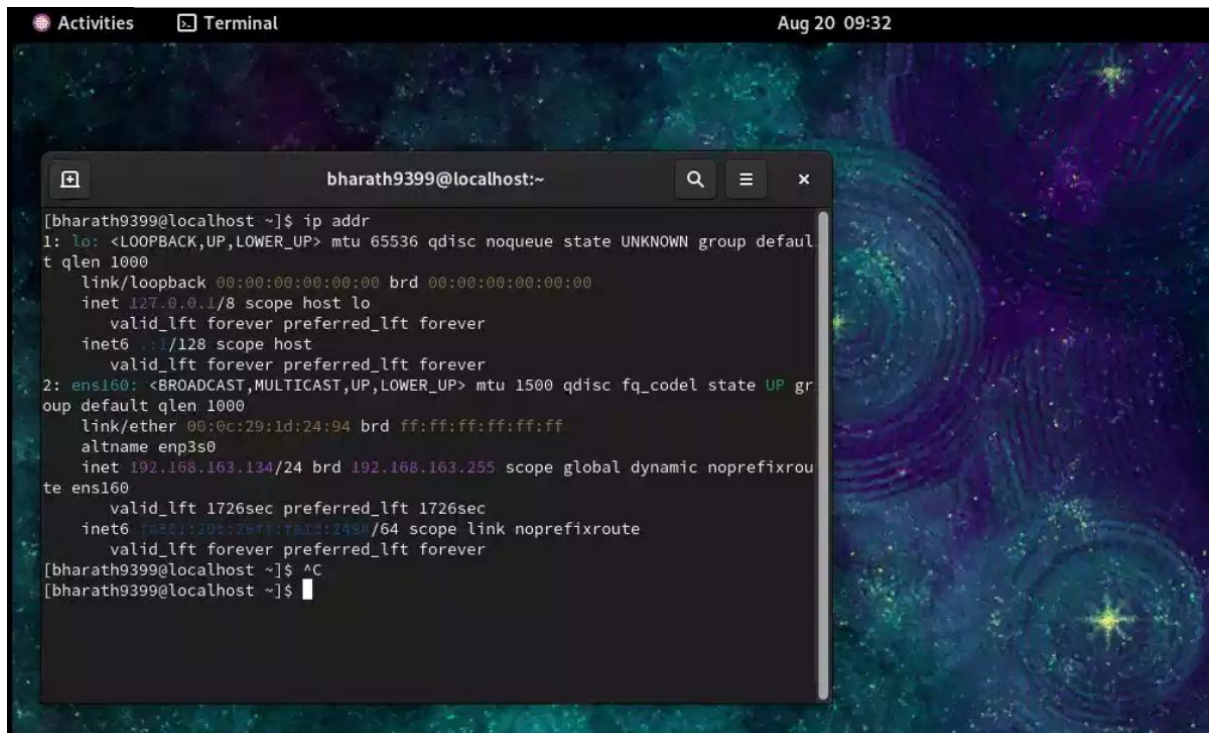
- history – Shows previously executed commands.

```
1  uname -a
2  hostname
3  uptime
4  dmesg
5  uptime
6  exit
7  uptime
8  dmesg
9  arch
10 date
11 whoami
12 uname
13 uname -r
14 uname -a
15 cal
16 last reboot
17 timedatectl
18 w
19 ps -t tty2
20 lshw
21 lsblk
22 df -h
23 df -i
24 du -s
25 du -sh
26 du -sh /var/log
27 du -sh /home
28 du -sh /home/*
29 su -sh /documents
30 su -sh /etc
31 du -sh /etc
32 ls
33 pwd
34 fdisk -l
35 fdisk -i
36 fdisk -|
37 fdisk -l
38 sudo su
39 history
```

```
1  pwd
2  fdisk -l
3  history
4  who
5  w
6  exit
7  pvs
8  vgs
9  lvs
10 pvdisplay
11 vgdisplay
12 lvdisplay
13 history
```

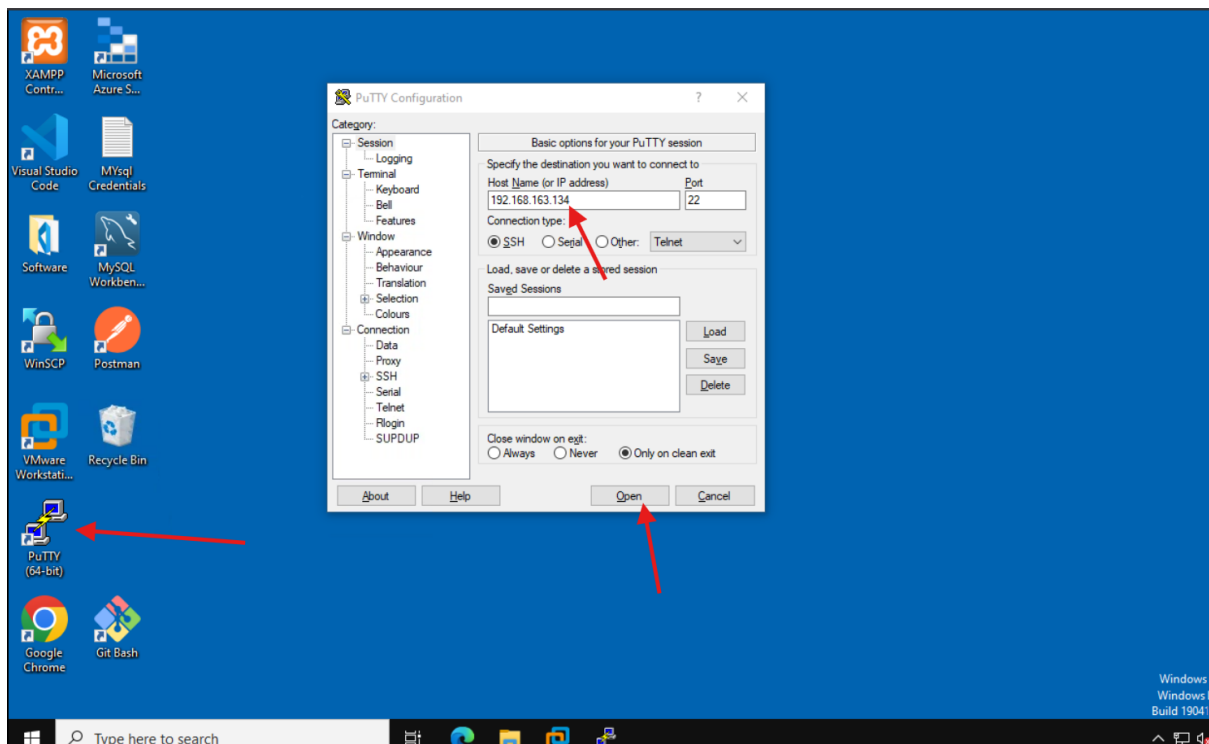
We'll now see how to access command line through putty see your IP using ip addr command

- ip addr - Shows network interfaces, their IP addresses, MAC addresses, and status. Replaces ifconfig.
- ifconfig - Old command to show **network interfaces and IPs**.

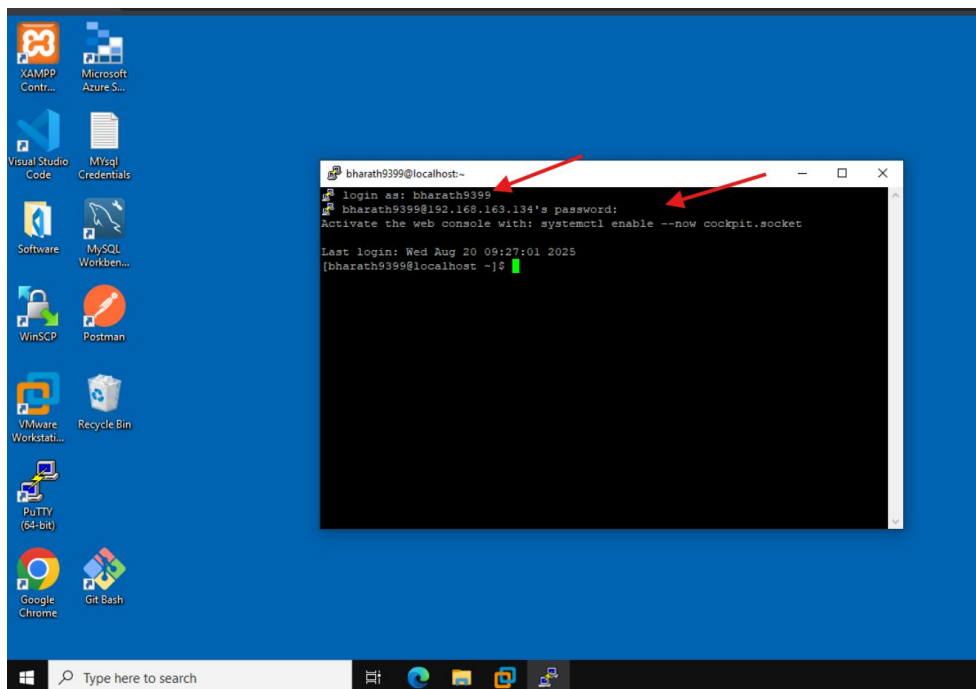


A terminal window titled "bharath9399@localhost:~" showing the output of the 'ip addr' command. The output displays details for the loopback interface 'lo' and the ethernet interface 'ens160'. The 'lo' interface has an IP address of 127.0.0.1. The 'ens160' interface has an IP address of 192.168.163.134. The terminal window is set against a dark background with a colorful, abstract pattern.

```
bharath9399@localhost:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:1d:24:94 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.163.134/24 brd 192.168.163.255 scope global dynamic noprefixroute ens160
        valid_lft 1726sec preferred_lft 1726sec
    inet6 fe80::20c:29ff:fe1d:2494/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
bharath9399@localhost:~]$ ^C
bharath9399@localhost:~$
```



Use the Display name of your Cent OS username and password



We use these commands now

- `ls -lh` - Lists files with permissions, owners, size, and timestamps. The `h` makes sizes human-readable (KB, MB, GB).
- `ls -a` - Shows hidden files.
- `ls -R` - Shows all files recursively, including those inside subdirectories.
- `ls -i` - Shows the inode number for each file (inode = low-level file info on disk).
- `ls -lt` - Lists files sorted by modification time, newest first.

```

bharath9399@localhost:~/Desktop
login as: bharath9399
bharath9399@192.168.163.134's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Wed Aug 20 09:27:01 2025
[bharath9399@localhost ~]$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:1d:24:94 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.163.134/24 brd 192.168.163.255 scope global dynamic noprefixroute ens160
        valid_lft 1032sec preferred_lft 1032sec
    inet6 fe80::20c:29ff:fe1d:2494/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[bharath9399@localhost ~]$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos
[bharath9399@localhost ~]$ ls -la
. .bash_history .bash_profile .cache Desktop Downloads .mozilla Pictures Templates
.. .bash_logout .bashrc .config Documents .local Music Public Videos
[bharath9399@localhost ~]$ pwd
/home/bharath9399
[bharath9399@localhost ~]$ ls -lh
total 0
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Desktop
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Documents
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Downloads
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Music
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Pictures
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Public
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Templates
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Videos
[bharath9399@localhost ~]$ cd
[bharath9399@localhost ~]$ cd desktop
-bash: cd: desktop: No such file or directory
[bharath9399@localhost ~]$ cd /Desktop
-bash: cd: /Desktop: No such file or directory
[bharath9399@localhost ~]$ cd Desktop
[bharath9399@localhost Desktop]$

```

- cd /etc - The /etc directory contains system configuration files (like network configs, user accounts, service settings). It's one of the most important folders in Linux.

```

[bharath9399@localhost ~]$ cd Desktop
[bharath9399@localhost Desktop]$ cd
[bharath9399@localhost ~]$ pwd
/home/bharath9399
[bharath9399@localhost ~]$ cd /etc
[bharath9399@localhost etc]$ ls
accounts.service  cron.hourly  firefox  issue  mailcap  makedumpfile.conf.sample  pbm2ppa.conf  sane.d  systemd
adjtime  cron.monthly  firewallld  issue.d  man_db.conf  pinforc  sasl2  system-release
aliases  cronstab  flatpak  issue.net  man_db.conf  pinforc  security  system-release-ope
alsa  cron.weekly  fonts  kdump  mclog  pkgconf  selinux  terminfo
alternatives  crypto-policies  foomatic  kdump.conf  microcode_ctl  pki  services  tmpfiles.d
anacronstab  crypttab  fprintd.conf  kernel  mime.types  plymouth  sestatus.conf  upstart-iss
appstream.conf  csh.cshrc  fstab  keys  mk2fs.conf  pm  setroubleshoot  trusted-key.key
asound.conf  csh.login  fuse.conf  keyutils  modprobe.d  pnm2ppa.conf  sgml  tuned
at.deny  cups  fwupd  krb5.conf  modules-load.d  polkit-1  udev
audit  cupshelpers  gdm  krb5.conf.d  motd  popt.d  shadow
authselect  dbus-1  gdm  ld.so.cache  motd.d  printcap  shadow-
avahi  dconf  geoclue  ld.so.conf  mtab  profile  shells
bash_completion.d  debuginfod  glvnd  ld.so.conf.d  multipath  profile.d  skel
bashrc  default  libaudit.conf  nanorc  protocols  smartmontools  usb_modeswitch.conf
bindresvport.blacklist  depmod.d  gnupg  libblockdev  netconfig  soc  vconsole.conf
binfmt.d  dhcpc  groff  libbtrfsd  NetworkManager  speech-dispatcher  vimrc
bluetooth  DIR_COLORS  group  libnl  networks  ssh  vmware-tools
brlapi.key  DIR_COLORS.lightbgcolor  group  libpaper.d  nftables  rsd  vulkan
brltty  dnf  grub2.cfg  libreport  nsswitch.conf  rc.d  statetab.d  wgetrc
brltty.conf  dnsmasq.conf  grub.d  libssh  nsswitch.conf.bak  rc.local  subgid  wireplumber
centos-release  dnsmasq.d  gshadow  libuser.conf  nvme  request-key.conf  subgid-
chromium  dracut.conf  gshadow-  locale.conf  openldap  request-key.d  subuid
chrony.conf  dracut.conf.d  gss  localtime  opt  resolv.conf  subuid-
chrony.keys  egl  host.conf  login.defs  os-release  sudo.conf  xdg
cifs-utils  enscript.cfg  hostname  logrotate.conf  ostree  rpm  sudoers  xml
cockpit  environment  hosts  logrotate.d  PackageKit  rpm  sudoers.d  yum
containers  ethertypes  hp  lsm  pam.d  rsyslog.conf  sudo-ldap.conf  yum.conf
cron.d  exports  inittab  lvm  papernize  rsyslog.d  sysconfig  yum.repos.d
cron.daily  favicon.png  inputrc  machine-id  passwd-  samba  sysctl.conf  sysctl.d
cron.deny  filesystems  iscsi  magic  passwd-  samba  sysctl.d
[bharath9399@localhost etc]$

```

```
bharath9399@localhost:~$ cd
bharath9399@localhost:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos
bharath9399@localhost:~$ ls -lt
total 0
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Desktop
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Documents
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Downloads
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Music
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Pictures
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Public
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Templates
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Videos
bharath9399@localhost:~$ ls -R
.:
Desktop Documents Downloads Music Pictures Public Templates Videos
./Desktop:
./Documents:
./Downloads:
./Music:
./Pictures:
./Public:
./Templates:
./Videos:
bharath9399@localhost:~$ ls -li
50331780 Desktop 50331781 Documents      139 Downloads      140 Music  16777348 Pictures 33685636 Public  16777347 Templates 33685637 Videos
bharath9399@localhost:~$ ls -la
total 28
drwx-----. 14 bharath9399 bharath9399 4096 Aug 19 14:48 .
drwxr-xr-x.  3 root      root      25 Aug 19 14:22 ..
-rw-----.  1 bharath9399 bharath9399  48 Aug 20 01:14 .bash_history
-rw-r--r--.  1 bharath9399 bharath9399  18 Feb 15 2024 .bash_logout
-rw-r--r--.  1 bharath9399 bharath9399 141 Feb 15 2024 .bash_profile
-rw-r--r--.  1 bharath9399 bharath9399 492 Feb 15 2024 .bashrc
drwx-----.  9 bharath9399 bharath9399 4096 Aug 19 14:23 .cache
```

- ls -lhs - Lists files sorted by size.
- ls -la - Shows all files (including hidden) with details.

```
root@localhost:/home/bharath9399
bharath9399@localhost:~$ sudo su
[sudo] password for bharath9399:
Sorry, try again.
[sudo] password for bharath9399:
Sorry, try again.
[sudo] password for bharath9399:
[root@localhost bharath9399]# ls
Desktop Documents Downloads Music Pictures Public Templates Videos
[root@localhost bharath9399]# ls -lhs
total 0
0 drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Desktop
0 drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Documents
0 drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Downloads
0 drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Music
0 drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Pictures
0 drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Public
0 drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Templates
0 drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Videos
[root@localhost bharath9399]# ls -la
total 28
drwx-----. 14 bharath9399 bharath9399 4096 Aug 19 14:48 .
drwxr-xr-x.  3 root      root      25 Aug 19 14:22 ..
-rw-----.  1 bharath9399 bharath9399  48 Aug 20 01:14 .bash_history
-rw-r--r--.  1 bharath9399 bharath9399  18 Feb 15 2024 .bash_logout
-rw-r--r--.  1 bharath9399 bharath9399 141 Feb 15 2024 .bash_profile
-rw-r--r--.  1 bharath9399 bharath9399 492 Feb 15 2024 .bashrc
drwx-----.  9 bharath9399 bharath9399 4096 Aug 19 14:23 .cache
drwxr-xr-x.  9 bharath9399 bharath9399 4096 Aug 20 01:14 .config
drwxr-xr-x.  2 bharath9399 bharath9399  6 Aug 19 14:22 Desktop
drwxr-xr-x.  2 bharath9399 bharath9399  6 Aug 19 14:22 Documents
drwxr-xr-x.  2 bharath9399 bharath9399  6 Aug 19 14:22 Downloads
drwx-----.  4 bharath9399 bharath9399  32 Aug 19 14:22 .local
drwxr-xr-x.  4 bharath9399 bharath9399  39 Aug 19 13:52 .mozilla
drwxr-xr-x.  2 bharath9399 bharath9399  6 Aug 19 14:22 Music
drwxr-xr-x.  2 bharath9399 bharath9399  6 Aug 19 14:22 Pictures
drwxr-xr-x.  2 bharath9399 bharath9399  6 Aug 19 14:22 Public
drwxr-xr-x.  2 bharath9399 bharath9399  6 Aug 19 14:22 Templates
drwxr-xr-x.  2 bharath9399 bharath9399  6 Aug 19 14:22 Videos
[root@localhost bharath9399]# hostname -i
::1 127.0.0.1
```

```
[bharath9399@localhost ~]$ ifconfig
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.163.134 netmask 255.255.255.0 broadcast 192.168.163.255
    inet6 fe80::20c:29ff:feld:2494 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:1d:24:94 txqueuelen 1000 (Ethernet)
    RX packets 2325 bytes 224379 (219.1 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1188 bytes 148421 (144.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 29 bytes 2813 (2.7 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 29 bytes 2813 (2.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

- ipconfig - command to show **network interfaces and ips**
- ping 192.168.163.134 - Sends test packets to IP 192.168.163.134 to check if it's reachable.

```
[bharath9399@localhost ~]$ ping 192.168.163.134
PING 192.168.163.134 (192.168.163.134) 56(84) bytes of data.
64 bytes from 192.168.163.134: icmp_seq=1 ttl=64 time=0.037 ms
64 bytes from 192.168.163.134: icmp_seq=2 ttl=64 time=0.264 ms
64 bytes from 192.168.163.134: icmp_seq=3 ttl=64 time=0.074 ms
64 bytes from 192.168.163.134: icmp_seq=4 ttl=64 time=0.097 ms
64 bytes from 192.168.163.134: icmp_seq=5 ttl=64 time=0.073 ms
64 bytes from 192.168.163.134: icmp_seq=6 ttl=64 time=0.060 ms
64 bytes from 192.168.163.134: icmp_seq=7 ttl=64 time=0.073 ms
64 bytes from 192.168.163.134: icmp_seq=8 ttl=64 time=0.097 ms
64 bytes from 192.168.163.134: icmp_seq=9 ttl=64 time=0.072 ms
64 bytes from 192.168.163.134: icmp_seq=10 ttl=64 time=0.075 ms
64 bytes from 192.168.163.134: icmp_seq=11 ttl=64 time=0.066 ms
64 bytes from 192.168.163.134: icmp_seq=12 ttl=64 time=0.072 ms

64 bytes from 192.168.163.134: icmp_seq=13 ttl=64 time=0.079 ms
64 bytes from 192.168.163.134: icmp_seq=14 ttl=64 time=0.081 ms
64 bytes from 192.168.163.134: icmp_seq=15 ttl=64 time=0.081 ms
64 bytes from 192.168.163.134: icmp_seq=16 ttl=64 time=0.082 ms
64 bytes from 192.168.163.134: icmp_seq=17 ttl=64 time=0.072 ms
64 bytes from 192.168.163.134: icmp_seq=18 ttl=64 time=0.079 ms
64 bytes from 192.168.163.134: icmp_seq=19 ttl=64 time=0.073 ms
64 bytes from 192.168.163.134: icmp_seq=20 ttl=64 time=0.081 ms
64 bytes from 192.168.163.134: icmp_seq=21 ttl=64 time=0.076 ms
^C
--- 192.168.163.134 ping statistics ---
21 packets transmitted, 21 received, 0% packet loss, time 20498ms
rtt min/avg/max/mdev = 0.037/0.084/0.264/0.041 ms
[bharath9399@localhost ~]$
```



Type here to search



- `sudo yum install whois -y` - installs the whois command-line tool without asking for confirmation.
- `whois` - The name of the package (a tool to fetch domain registration info).
- `yum` - Package manager for **RHEL, CentOS, Fedora** systems. It installs, updates, and removes software.

```
[bharath9399@localhost ~]$ sudo yum install whois -y
[sudo] password for bharath9399:
Last metadata expiration check: 1:46:24 ago on Wednesday 20 August 2025 09:54:29 AM.
Dependencies resolved.
=====
Package                                Architecture                          Version
=====
Installing:
whois                                  x86_64                               5.5.9-4.el9
Installing dependencies:
whois-libs                             noarch                               5.5.9-4.el9
=====
Transaction Summary
=====
Install 2 Packages

Total download size: 110 k
Installed size: 348 k
Downloading Packages:
(1/2): whois-libs-5.5.9-4.el9.noarch.rpm
(2/2): whois-5.5.9-4.el9.x86_64.rpm
=====
Total
CentOS Stream 9 - AppStream
```

- `whois gmail.com` - Shows **domain registration info** (owner, registrar, dates).
- `whois wipro.com` - Shows **domain registration info** (owner, registrar, dates).

```
[bharath9399@localhost ~]$ whois gmail.com
Domain Name: GMAIL.COM
Registry Domain ID: 4667231_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2025-07-11T10:10:56Z
Creation Date: 1995-08-13T04:00:00Z
Registry Expiry Date: 2026-08-12T04:00:00Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
Name Server: NS1.GOOGLE.COM
Name Server: NS2.GOOGLE.COM
Name Server: NS3.GOOGLE.COM
Name Server: NS4.GOOGLE.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2025-08-20T06:13:26Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
```

- `dig wipro.com` → Performs a **DNS lookup** for the domain, showing IPs, name servers, etc.
- `nslookup wipro.com` → Simple **DNS query tool** to resolve domain names.


```
[bharath9399@localhost ~]$ whois wipro.com
Domain Name: WIPRO.COM
Registry Domain ID: 2559888_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.registrar.amazon
Registrar URL: http://registrar.amazon.com
Updated Date: 2023-01-11T06:32:33Z
Creation Date: 1992-12-08T05:00:00Z
Registry Expiry Date: 2030-12-07T05:00:00Z
Registrar: Amazon Registrar, Inc.
Registrar IANA ID: 468
Registrar Abuse Contact Email: trustandsafety@support.aws.com
Registrar Abuse Contact Phone: +1.2024422253
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: NS-1145.AWSDNS-15.ORG
Name Server: NS-1575.AWSDNS-04.CO.UK
Name Server: NS-326.AWSDNS-40.COM
Name Server: NS-659.AWSDNS-18.NET
DNSSEC: signedDelegation
DNSSEC DS Data: 31918 13 2 0074B6D159450F1E12E2243EFEA93235A69CBDB77A1CD6A795EEA4F87E7A5C9E
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2025-08-20T06:14:27Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
```

```
[bharath9399@localhost ~]$ dig wipro.com

; <<>> DiG 9.16.23-RH <<>> wipro.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 20757
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 4096
;; QUESTION SECTION:
;wipro.com.                IN      A

;; ANSWER SECTION:
wipro.com.                5       IN      A      18.205.78.10

;; Query time: 72 msec
;; SERVER: 192.168.163.2#53(192.168.163.2)
;; WHEN: Wed Aug 20 11:46:47 IST 2025
;; MSG SIZE rcvd: 54

[bharath9399@localhost ~]$
```

```
[bharath9399@localhost ~]$ nslookup wipro.com
Server:          192.168.163.2
Address:         192.168.163.2#53

Non-authoritative answer:
Name:   wipro.com
Address: 18.205.78.10
```

- `netstat -tuln` → Shows listening services and open ports on your system.
- `netstat -pnltu` → Same as above, but also shows process IDs.

```
[bharath9399@localhost ~]$ netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN
tcp6       0      0 :::22                   :::*                    LISTEN
tcp6       0      0 :::1:631                :::*                    LISTEN
udp        0      0 0.0.0.0:5353            0.0.0.0:*               LISTEN
udp        0      0 127.0.0.1:323           0.0.0.0:*               LISTEN
udp        0      0 0.0.0.0:33783           0.0.0.0:*               LISTEN
udp6       0      0 :::5353                  :::*                    LISTEN
udp6       0      0 :::37003                  :::*                    LISTEN
udp6       0      0 :::1:323                  :::*                    LISTEN
[bharath9399@localhost ~]$ netstat -pnltu
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN      -
tcp6       0      0 :::22                   :::*                    LISTEN      -
tcp6       0      0 :::1:631                :::*                    LISTEN      -
udp        0      0 0.0.0.0:5353            0.0.0.0:*               LISTEN      -
udp        0      0 127.0.0.1:323           0.0.0.0:*               LISTEN      -
udp        0      0 0.0.0.0:33783           0.0.0.0:*               LISTEN      -
udp6       0      0 :::5353                  :::*                    LISTEN      -
udp6       0      0 :::37003                  :::*                    LISTEN      -
udp6       0      0 :::1:323                  :::*                    LISTEN      -
[bharath9399@localhost ~]$
```

```
[bharath9399@localhost ~]$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
[bharath9399@localhost ~]$ ls -lh
total 0
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Desktop
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Documents
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Downloads
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Music
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Pictures
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Public
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Templates
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Videos
[bharath9399@localhost ~]$ mkdir Batch8
[bharath9399@localhost ~]$ ls
Batch8 Desktop Documents Downloads Music Pictures Public Templates Videos
[bharath9399@localhost ~]$ ls -la
total 28
drwx-----. 15 bharath9399 bharath9399 4096 Aug 20 11:49 .
drwxr-xr-x.  3 root          root          25 Aug 19 14:22 ..
-rw-----.  1 bharath9399 bharath9399   48 Aug 20 01:14 .bash_history
-rw-r--r--.  1 bharath9399 bharath9399  18 Feb 15 2024 .bash_logout
-rw-r--r--.  1 bharath9399 bharath9399 141 Feb 15 2024 .bash_profile
-rw-r--r--.  1 bharath9399 bharath9399 492 Feb 15 2024 .bashrc
drwxr-xr-x.  2 bharath9399 bharath9399    6 Aug 20 11:49 Batch8
drwx-----.  9 bharath9399 bharath9399 4096 Aug 19 14:23 .cache
drwxr-xr-x.  9 bharath9399 bharath9399 4096 Aug 20 01:14 .config
drwxr-xr-x.  2 bharath9399 bharath9399    6 Aug 19 14:22 Desktop
drwxr-xr-x.  2 bharath9399 bharath9399    6 Aug 19 14:22 Documents
drwxr-xr-x.  2 bharath9399 bharath9399    6 Aug 19 14:22 Downloads
```

- mkdir Batch8 → Creates a new folder named Batch8 in current directory.
- mkdir -p Renu/Wipro/test → Creates nested folders (Renu, inside it Wipro, inside it test) all at once.

```
[bharath9399@localhost ~]$ mkdir -p Bharath/wipro/test
[bharath9399@localhost ~]$ ls -lh
total 0
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 20 11:49 Batch8
drwxr-xr-x. 3 bharath9399 bharath9399 19 Aug 20 11:51 Bharath
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Desktop
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Documents
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Downloads
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Music
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Pictures
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Public
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Templates
drwxr-xr-x. 2 bharath9399 bharath9399 6 Aug 19 14:22 Videos
[bharath9399@localhost ~]$ cd Bharath
[bharath9399@localhost Bharath]$ ls
wipro
[bharath9399@localhost Bharath]$ cd wipro
[bharath9399@localhost wipro]$ ls
test
```

```
[bharath9399@localhost wipro]$ cd
[bharath9399@localhost ~]$ pwd
/home/bharath9399
[bharath9399@localhost ~]$ ls
Batch8  Bharath  Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
[bharath9399@localhost ~]$ cd Bharath
[bharath9399@localhost Bharath]$ ls
wipro
[bharath9399@localhost Bharath]$ cd wipro
[bharath9399@localhost wipro]$ ls
test
[bharath9399@localhost wipro]$ cd
[bharath9399@localhost ~]$
```

- cd Renu/ → Moves into a user-created folder named Bharath in the current directory.
- cd Wipro/ → Moves into Wipro folder (inside Renu, based on your path).
- cd test/ → Goes into a folder named test (created under Bharath/Wipro).
- cd Bharath/wipro/test/ → Moves into the nested directory structure.
- cd .. → Goes one level up (for example, from /home/user/Bharath/wipro/test to /home/user/Bharath/wipro).

```
renu@localhost:~  
3 ip addr  
4 ls  
5 sudo su  
6 ls -a  
7 pwd  
8 ls -lh  
9 cd  
10 cd /Desktop  
11 cd Desktop  
12 cd  
13 cd /etc  
14 ls  
15 ls -lh  
16 cd  
17 ls  
18 cd /etc  
19 ls -l  
20 ls -lt  
21 ls -R  
22 ls -i  
23 ls -la  
24 ls -lhs  
25 cd
```

```
renu@localhost:~  
24 ls -lhs  
25 cd  
26 hostname -l  
27 hostname -| exit  
28 ip addr  
29 ifconfig  
30 ip addr  
31 ifconfig  
32 ping 192.168.163.133  
33 whois gmail.com  
34 sudo yum install whois -y  
35 whois gmail.com  
36 whois wipro.com  
37 dig wipro.com  
38 dig omprakas.me  
39 netstat -tuln  
40 netstat -pnltu  
41 nslookup wipro.com  
42 ls  
43 ls -lh  
44 mkdir Batch8  
45 ls -lh  
46 mkdir -p Renu/Wipro/test
```

```
renu@localhost:~  
47 ls -lh  
48 cd Renu/  
49 ls  
50 cd Wipro/  
51 ls  
52 cd test/  
53 ls  
54 cd  
55 touch batch8.txt  
56 ls  
57 ls -lh  
58 cd Batch8/  
59 ls  
60 cd  
61 cd Renu/  
62 ls  
63 cd  
64 cd Renu/Wipro/test/  
65 cd..  
66 cd ..  
67 cd  
68 history  
[renu@localhost ~]$
```